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Department of Artificial Intelligence and Data Science

Innovative Teaching Practice -Flipped Classroom Methodology

Subject Code&Name: AL3451-Machine Learning

Topic: Guidelines for Conducting Machine Learning Experiments

Methodology: Flipped Classroom

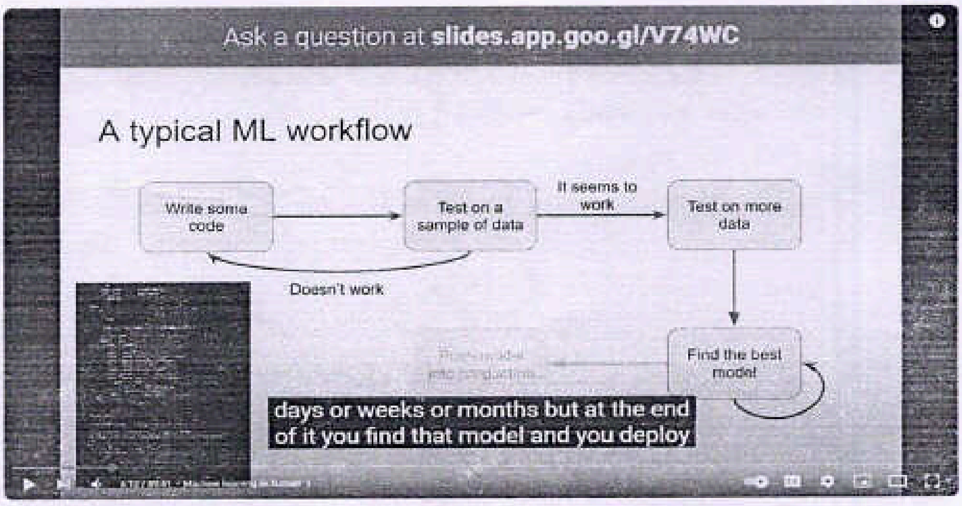
Objective: Equip students with foundational knowledge of how to plan and execute machine learning experiments.

Topics Covered:

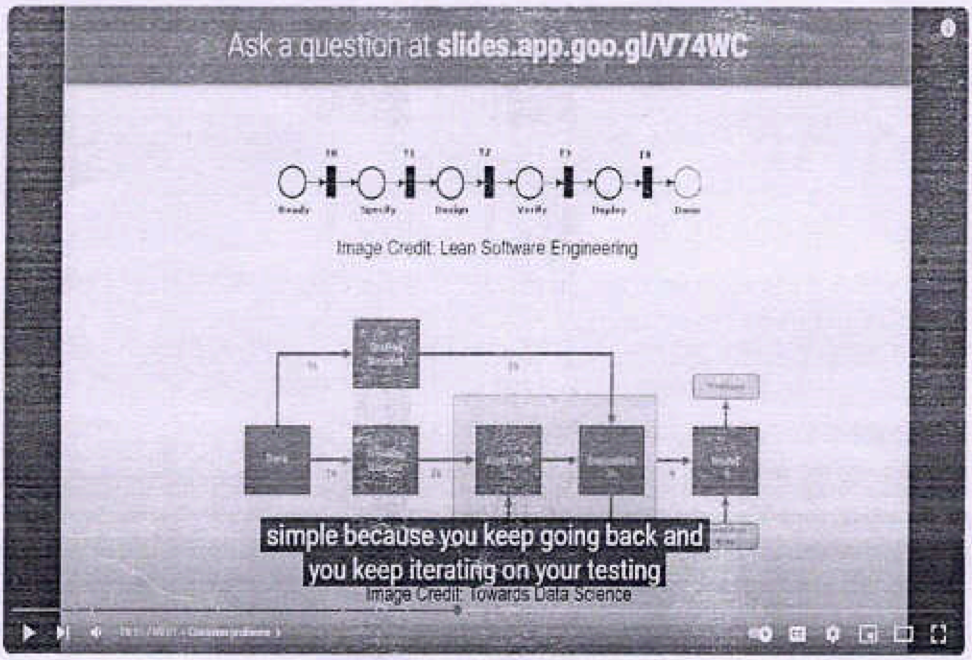
- Problem definition and hypothesis
- Data preparation and splitting (training, validation, test)
- Model selection and justification
- Evaluation metrics (accuracy, F1-score, etc.)
- Reproducibility practices (version control, seed setting)
- Error analysis and iterative improvement
- Reporting results

Learning Materials Provided:

- Video lecture:

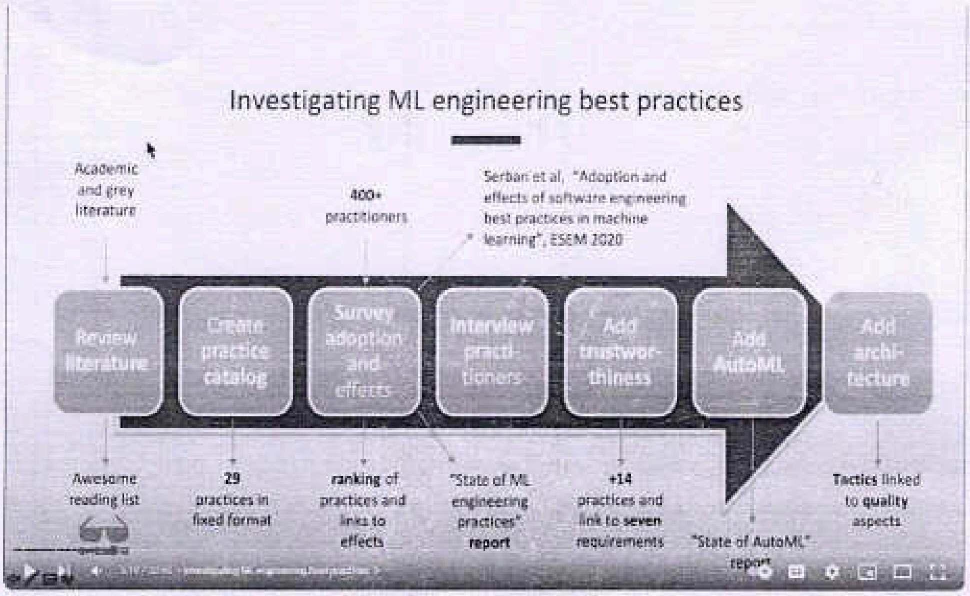


Experiment Management for Machine Learning

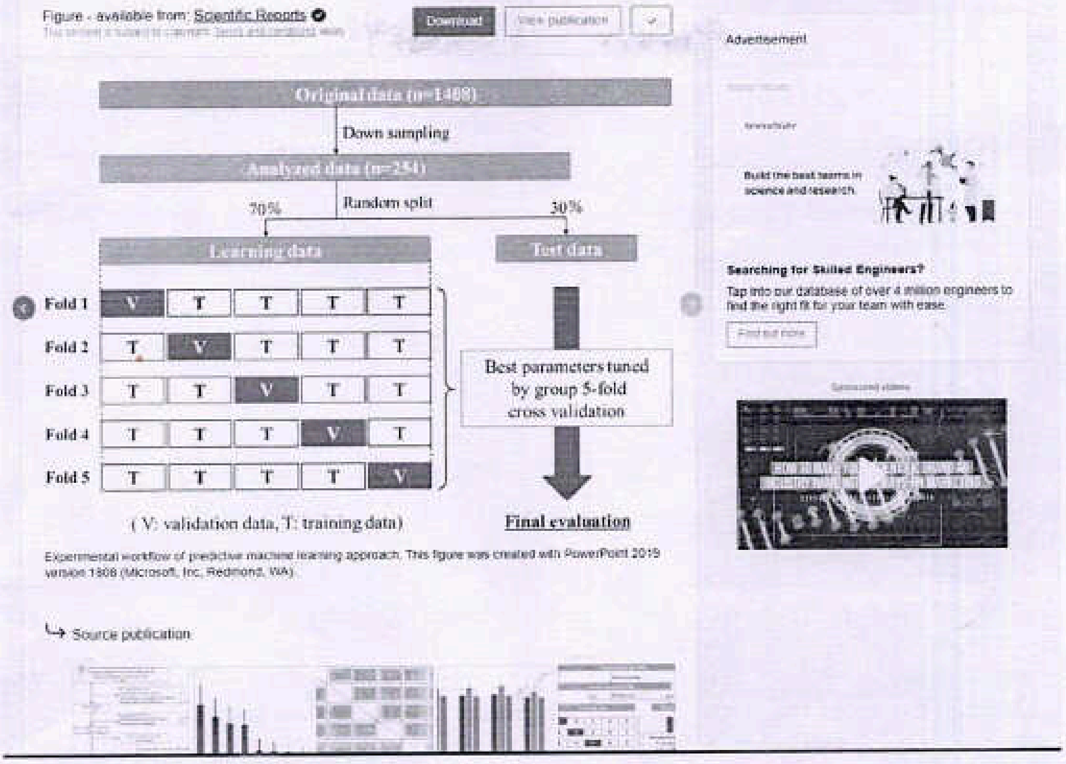


Experiment Management for Machine Learning

- "Research paper: *"Reproducible Machine Learning: Guidelines and Best Practices"*



Engineering Best Practices for Machine Learning // Alex Serban // MLOps Meetup #79



Quiz: 5 questions to check basic understanding

Quiz: Basic Understanding of ML Experiment Guidelines

1. Which of the following is the correct order of steps in a typical machine learning experiment?

A. Model training → Data cleaning → Evaluation
B. Data collection → Preprocessing → Model training → Evaluation → Reporting
C. Evaluation → Data collection → Model deployment
D. Model deployment → Evaluation → Data labeling

Correct Answer: **B**

2. Why is it important to split the dataset into training, validation, and test sets?

A. To reduce the amount of data used in the experiment
B. To ensure the model memorizes the data
C. To fairly evaluate model performance and avoid overfitting
D. To make the code easier to write

Correct Answer: **C**

3. What does setting a random seed in an ML experiment help ensure?

A. Faster training speed
B. Better accuracy
C. Reproducibility of results
D. Larger datasets

Correct Answer: **C**

4. Which metric would be most appropriate to evaluate a model on an imbalanced classification dataset?

A. Accuracy
B. F1-score
C. Mean squared error
D. Precision alone

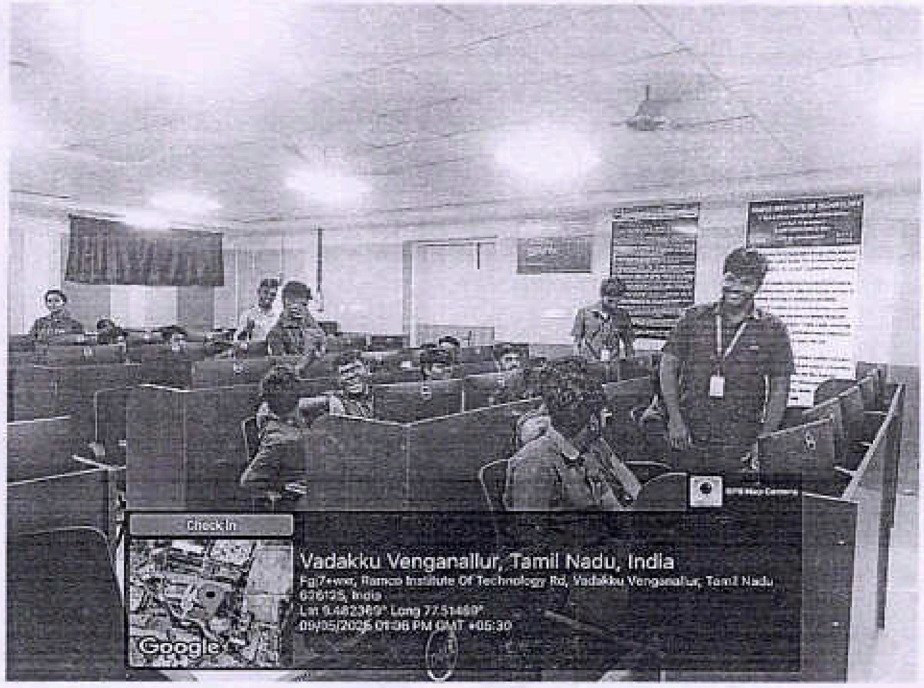
Correct Answer: **B**

5. What is the primary purpose of error analysis in an ML experiment?

A. To make the experiment faster
B. To debug the code
C. To understand model weaknesses and improve performance
D. To increase the size of the training data

Correct Answer: **C**

<u>**Screenshot showing active participation of the students**</u>



Faculty in-charge 19/5/25

Dr.R.M.Rajeshwari

19/5

HOD

Dr.M.Kaliappan

Department of Artificial Intelligence and Data science

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: B.Tech/IV/AD

Course Code & Title:AL3451- Machine Learning

Name of the Faculty member (s): Dr.R.M.Rajeshwari

Innovative Practice Description

- **Unit / Topic: Unit 5 / Guidelines for machine learning experiments**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO40
- **Activity Chosen: Flipped Class room**
- **Justification:**
- The flipped classroom enhances understanding by allowing students to learn theoretical concepts at their own pace and apply them actively during class. It promotes critical thinking, hands-on experience, and collaboration through real-time problem-solving. This approach helps students design and conduct effective machine learning experiments more confidently and effectively

Time Allotted for the Activity: 20 minutes

- **Details of the Implementation:**

In a flipped classroom approach, the implementation begins with **pre-class activities**, where students are provided with pre-recorded video lectures, tutorials, and reading materials covering core machine learning concepts such as data preprocessing techniques, model selection strategies, and evaluation metrics. These resources are hosted on platforms like an LMS (e.g., Moodle or Google Classroom), allowing students to study at their own pace. To ensure engagement, quizzes and reflection questions are included to test their understanding before the class session.During the **in-class activities**, the focus shifts to applying the pre-learned concepts. The session starts with a brief recap and Q&A to clarify any doubts. Students then engage in hands-on coding sessions where they work on real datasets—cleaning data, building models using tools like scikit-learn or Tensor Flow, and evaluating model performance. This is followed by collaborative group activities, where students tackle case studies in domains such as healthcare or agriculture, design ML pipelines, and solve practical problems. Peer discussions and feedback sessions are included to encourage collaborative learning and improve critical thinking.After the classroom session, **post-class activities** are used to reinforce learning. Students may be asked to write reflections, submit a short video explanation, or complete assignments that require them to apply what they've learned to a new dataset. Additional quizzes or short tests help assess how well they've grasped the material and ensure continued engagement with the topic.

Finally, **assessment and feedback** are integral to the process. Formative assessments occur throughout the classroom sessions via hands-on work and discussions, while summative assessments involve evaluating project work, assignments, and quizzes. Instructors provide timely feedback to guide students in refining their understanding and improving their skills in designing and conducting machine learning experiments.

- **CO – PO / PSO mapping:**

CO	PO4	PSO2
CO2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO4	PSO2
	2	2
	Justification for correlation	
PO4	The given topic aligns with PO4 as it emphasizes the application of research-based methodologies in machine learning. Understanding guidelines for designing and conducting experiments including data preprocessing, model selection, and evaluation and enables students to systematically validate the performance and effectiveness of ML algorithms, fulfilling the objective of PO4.	
PSO2	The given topic supports PSO2 as it equips students with foundational skills required to apply machine learning techniques effectively in various domains like Healthcare, Education, Agriculture, and more. By understanding data preprocessing, model selection, and evaluation strategies, students are prepared to design robust ML experiments that can accurately forecast future events and trends in these critical sectors.	

- **Images / Screenshot of the practice:**

- Reflective Critique:
  - *Feedback of practice from students and other stakeholders:*
  - Concepts were understood more deeply by learning at one's own pace before class and applying them through hands-on practice during sessions.
  - The learning process was made more engaging and practical by working on real datasets and collaborating with peers.
  - Confidence in applying machine learning techniques was gained by using class time for coding and problem-solving instead of lectures.

- ***Benefit of the practice:*** (**E.g.: Outcome attainment would have increased due to innovative practice over conventional practice**)

  - Students gain practical, research-based experience in ML experimentation.
  - Improved understanding of preprocessing, model choice, and evaluation.
  - Enhanced problem-solving, collaboration, and critical thinking skills.

Challenges faced in implementation:

Implementing a flipped classroom for machine learning experiment design presents several important challenges. One major challenge is ensuring that all students come prepared by completing the pre-class materials, as lack of preparation can hinder active participation during hands-on sessions. Additionally, developing high-quality instructional content such as videos and reading materials demands significant time, effort, and technical resources from instructors. Managing the classroom effectively also poses difficulties, particularly when facilitating collaborative activities and providing individual support in large classes or among students with diverse skill levels. Addressing these challenges is crucial for the successful adoption of the flipped classroom model..

References:

- Lage, M. J., Platt, G. J., & Treglia, M. (2000). *Inverting the Classroom: A Gateway to Creating an Inclusive Learning Environment*. The Journal of Economic Education, 31(1), 30-43.
- https://www.deeplearning.ai/machine-learning-yearning/

19/5/25

Signature of Faculty Member

19/2/5

HOD